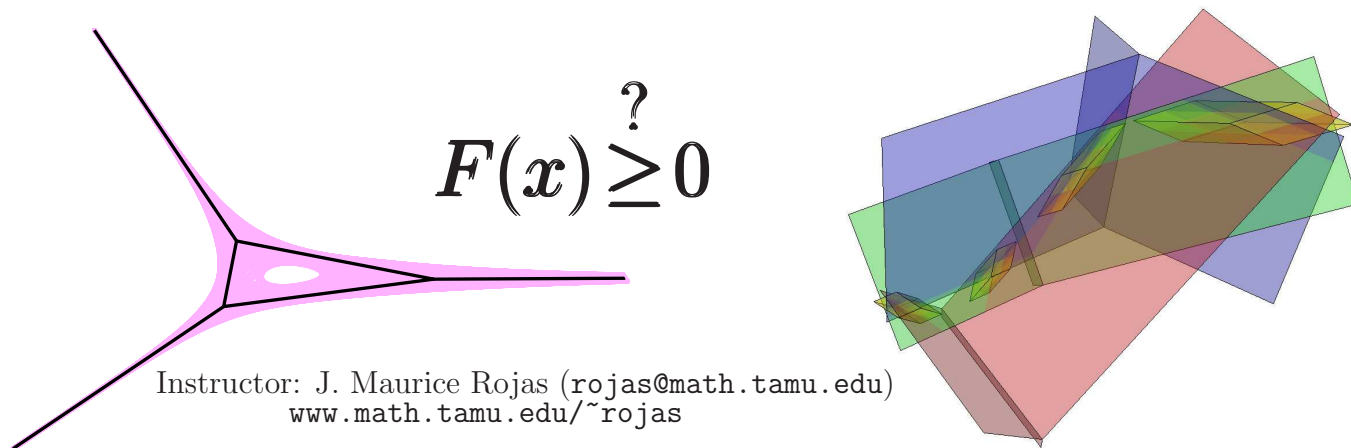


MATH 648 (Fall 2022), TuTh 5:30-6:45PM, Blocker TBA

INTRODUCTION TO ALGORITHMIC ALGEBRAIC GEOMETRY*



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Algebraic Geometry was born over two millenia ago, with the aim of understanding how to solve polynomial equations. Many important problems in engineering, computer science, chemistry, and physics are reducible to polynomial system solving. So there is a serious need to understand complexity issues, such as the most efficient way to solve a system of equations one may encounter in practice. Algebraic algorithms also motivate great mathematical ideas in many different areas.

GOALS Math 648 gives an introduction to algebraic geometry through modern algorithms for solving nonlinear systems of equations. Students completing this course will be well-prepared to understand recent developments in areas such as optimization, coding theory, algebraic complexity, algebraic statistics, algorithmic number theory, and computational algebra. This course also provides valuable motivation and background for more abstractly oriented courses on algebraic geometry.

PREREQUISITIES Math 653 (first semester graduate algebra) **or my consent**. In particular, we assume **no** background in commutative algebra or complexity theory: any necessary algebraic or algorithmic background will be developed as needed. Motivated upper division students in mathematics, computer science, and engineering will also find the material quite accessible.

SYLLABUS Topics vary from semester to semester, but usually include a large subset of the following: connections to linear and semidefinite programming, basics of amoebae and tropical geometry, homotopy inspired proof of the Fundamental Theorem of Algebra, Descartes' Rule of Signs and its generalizations, binomial systems, Newton's method, numerical conditioning, connections to polyhedra and toric Varieties, $\mathcal{A}$ -discriminants, Smale's 17th Problem, Alpha Theory and the Effective Inverse Function Theorem, Bernstein's Theorem, toric resultants, Turing/BSS Complexity and the $\mathbf{P} \stackrel{?}{=} \mathbf{NP}$ Question, Fewnomial Theory, polynomial identity testing and **BPP**, connections to number theory, connections to Gröbner Bases.

GRADING 50% biweekly homeworks, 50% project (including 20-30 minute presentation). Typically, students lecture during the last 2-3 weeks of the course. Projects on applications coming from **outside** of mathematics are highly encouraged.

* A more detailed description can be found at <http://www.math.tamu.edu/~rojas/648.html>